Graded Fault Response for Autonomous Driving via Continuous Sensor Health Monitoring and Risk/Capability-Aware MRM Supervision

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Abstract— Sensor-fault handling in automated driving stacks largely reduces to an immediate in-lane stop, while undetected degradation lets the vehicle keep driving with impaired perception. We propose a graded fault-response layer added onto an unmodified Autoware stack: (i) a behavior-tree monitor estimating per-sensor health as continuous scores (freshness, quality, physics) composed into capability scores, and (ii) an MRM supervisor selecting between shoulder/parking evacuation and deferral to the stock stop by fault type, severity, and remaining capability, with a conservative 3.0 m/s cap on evacuation movement. Across 1,862 valid closed-loop CARLA cases, detection reached 96–100% in every condition except one tolerated by design, with zero false activations, and degraded-state exposure fell from 238–275 m to 11–58 m. Verdicts rest on a dual scheme—an ego collision ledger and a pedestrian physics oracle—built after finding silent collision sensors. A pre-registered dose-response experiment (3 caps × 4 distances × 10 seeds) shows the cap causally moves the avoid/collide boundary inward (8 m: discordant pairs 10:0, exact McNemar $p = 0.002$); encounters in the post-detection deceleration transient remain at risk.

I. INTRODUCTION

Automated driving systems at SAE Level 4 and above cannot presuppose driver intervention: upon a fault they must reach a minimal risk condition (MRC) via a minimal risk maneuver (MRM) on their own. ISO 26262 requires detection and response within the fault-tolerant time interval (FTTI), and SOTIF (ISO 21448) requires managing hazards from sensor performance limitations [1], [2], [3]. In the open-source stack Autoware, the default MRMs are in-lane stops (comfortable/emergency), and its stock diagnostics primarily target communication- and hardware-level anomalies such as topic silence and timeouts [7]. In the version and execution configuration evaluated in this study (Autoware 0.39.0), degradation faults whose values stay within valid ranges left driving to continue without intervention, and no supervisory layer existed that evaluates fault type and degree together with remaining maneuvering capability, selects between evacuation and stopping, and limits evacuation speed to match the fault state. That supervisory layer—not the shoulder maneuver itself—is the contribution of this work; the observation is version-specific, not a claim about Autoware in general.

An in-lane stop is simple and immediate, but the stopped vehicle blocks traffic and creates secondary-collision exposure until the lane clears. Conversely, driving on with an undetected fault extends the distance exposed to hazards under degraded perception—in our unmonitored observations (one run per fault type) the stock stack completed 238–275 m of route with faults injected. The design space of fault response has three axes: whether to stop now, how far to move, and—if moving—how fast.

Automating human-driver-like shoulder evacuation requires (i) distinguishing faults mild enough to permit moving from faults requiring an immediate stop; (ii) a continuous estimate of fault type and degree (a binary normal/faulty state is insufficient); and (iii) managing the risk of moving itself in a degraded state—the stopping distance against a dart-out. We implement these as a continuous health score, risk/capability-based state transitions, and a conservative speed cap.

Finally, simulation-based safety claims stand only on the reliability of the collision verdict itself: we empirically uncovered a simulator collision-sensor configuration that remained silent, and built a dual adjudication scheme—a real-time ego collision-sensor ledger and a pedestrian physics oracle—on which every safety conclusion rests.

Contributions. (1) A behavior-tree monitor that scores per-sensor freshness, quality, and physical consistency independently and composes them over eight spatial sectors into perception/localization capability scores, published as continuous health. (2) A risk/capability-aware MRM supervisor that selects between shoulder/parking evacuation (MRM_MOVE) and deferral to the stock stop (MRM_DEFER) by health-score bands, chooses the refuge by a cost function over degraded-state exposure time, traffic-light observability, and sector deficits, and imposes a fixed 3.0 m/s cap (below the model-derived bound ≈3.9 m/s) on evacuation movement. (3) A dual collision-adjudication methodology (ego RAW-HIT ledger + pedestrian physics oracle) with a catalogue of verdict pitfalls in runner exit labels and official infraction records. (4) Validation totalling 1,862 valid closed-loop cases, including a pre-registered dose-response experiment verifying the causal cap effect on settled encounters, and transient-vs-settled observations locating residual evacuation risk in the post-detection transient.

The novelty is not in the individual techniques (CUSUM, residual monitoring, noisy-OR fusion, stopping-distance model—each known), but in the integrated supervisory layer spanning detection through selection and speed, and in the closed-loop demonstration whose verdict reliability is secured at the methodological level.

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II. RELATED WORK

Standards and fault models. Functional-safety and SOTIF frameworks [1], [2], [3] require runtime monitoring and responses to degradation; ISO 23793-1 [4] taxonomizes MRM types but covers only in-lane stops, and industrial ISO 3691-4 [5] prescribes "limited speed for limited capability," which we transfer to on-road MRM. None provides a way to continuously estimate fault kind and degree and select response and speed. Classical FDI—residual generation/decision [6] and sequential change detection (our LiDAR channel borrows CUSUM)—ends at detection. Our fault models follow measured LiDAR soiling [8], rain scattering [9], and GNSS correlated bias [10].

Fallback and MRM planning. Fail-safe trajectory planning [11] maintains stop trajectories and online verification but takes the fault assessment and response choice as given. Reschka and Maurer [12] noted that an in-lane stop is not always a safe state—the concern we start from. Closest to us, Xue et al. [13] presuppose a forward-sensor fault and plan safe-zone stops under MPC constraints, and Tang et al. [14] adaptively select MRM behaviors under multiple sensor faults via HAZOP risk analysis with TTC/THW evaluation. They differ in that (i) the fault state is presupposed rather than estimated online; (ii) selection uses surrounding-traffic TTC/THW, whereas we combine risk grace time with remaining perception/localization capability and limit the evacuation speed itself; (iii) validation checks representative scenarios, whereas we verify 1,862 cases on a real stack with methodologically secured encounter verdicts. Capability-based self-monitoring models abilities for online decisions [15]; our sector composition couples capability scores directly to MRM selection and the cap (behavior trees as the implementation structure).

Simulation-based evaluation. CARLA [16] hosts fault-injection studies that inject faults to find vulnerabilities—the opposite direction from ours, where the stack itself detects, grades, and responds. Our infrastructure builds on AWCL [17], the TUM Autoware–CARLA coupling. We found no prior report addressing simulated collision-verdict reliability at the methodology level.

III. PROPOSED SYSTEM

A. Architecture

The system consists of two independent add-on nodes (Fig. 1). The monitoring module inspects each sensor at 10 Hz on a behavior-tree basis, computing a per-sensor health score in $[0, 1]$ from three detector families: timeout (freshness), quality (noise, valid ratio), and physics (residual, bias). The MRM supervisor subscribes to the scores and runs a state machine: for evacuation it changes the route via the Autoware AD API and imposes a speed cap on the planning layer; for deferral it triggers the stock mrm_handler stop through the diagnostic graph. Maneuver execution is delegated to the stock stack; the main stack is not modified. Reception accumulates at native sensor rates; evaluation runs at 10 Hz (AUTOSAR watchdog-style periodic supervision).

B. Sensor Scoring

Freshness. For elapsed time $\mathrm{age}(t)$ since last reception, $s_{\mathrm{fresh}} = \max(0,\, 1 - \mathrm{age}/\tau_{\mathrm{detect}})$ decays linearly—the score can cross the downstream band boundary before the budget is exhausted, enabling early transition under silence; a k-miss rule penalizes early on $3\times$ the median reception interval.

Quality. For LiDAR, a noise level $\hat{\sigma} = \overline{|\Delta r|} \cdot \sqrt{\pi}/2$ (neighboring-point range differences, $|\Delta r| < 1$ m) is referenced to a boot-time baseline that refuses to track abrupt rises (the fault cannot become its own baseline); the score is $\exp(-\max(0, \hat{\sigma} - \mathrm{base})/0.1)$ times a valid-point-ratio score, with a parallel CUSUM test so detection delay is inversely proportional to fault magnitude. Camera quality applies the same structure on inter-frame pixel differences (no cumulative test: they rise with scene motion). Sparsification is caught by the valid-ratio channel, clutter by the range-discontinuity channel.

Localization (three channels). (a) a residual channel on the GNSS–EKF distance (sensitive to noise and correlated bias); (b) a bias channel comparing 3-s means of IMU and EKF yaw rates (normal turning cancels; gyro bias remains on the IMU side); (c) a silence channel decaying the score when GNSS input stops $\geq$1 s while driving—covering the residual channel's blind spot of freezing at the last value.

C. Sector and Capability Synthesis

Sensor scores map to eight directional sectors by mounting direction and field of view and fuse as $s_{\mathrm{sector}} = 1 - \prod_i (1 - s_i)$; perception capability is a front-weighted sector sum combined with traffic-light observability at 0.6:0.4; system health is $\min(\mathrm{perception}, \mathrm{cap}_{\mathrm{loc}})$—the weakest link.

D. Risk-Coupled Detection Budget

The freshness decay constant is coupled to driving risk: $\tau_{\mathrm{detect}}(t) = \min(\tau_{\mathrm{fresh}},\, \mathrm{TTC}_{\mathrm{cons}}(t))$, where $\mathrm{TTC}_{\mathrm{cons}}$ is the minimum of nearest-obstacle TTC and path-crossing time with a fail-conservative hold when the TTC source itself is unhealthy. The design intent is $T_{\mathrm{detect}} + T_{\mathrm{response}} + T_{\mathrm{margin}} \leq T_{\mathrm{hazard}}$; the implementation is a risk-adaptive heuristic approximation, not a formal guarantee (explicit FRTI/margin reservation is future work). For degradation faults the requirement is a severity curve $T_{\mathrm{detect}}(\sigma) \leq T_{\mathrm{hazard}}(\sigma)$.

E. MRM Supervisor

Dual thresholds on health (upper 0.7, lower 0.3) span three bands: NORMAL, evacuation (MRM_MOVE), and deferral (MRM_DEFER, on floor entry or localization-confidence collapse); decay constants leave a wide insensitive gap between faulted and normal scores, so behavior follows the band structure, not

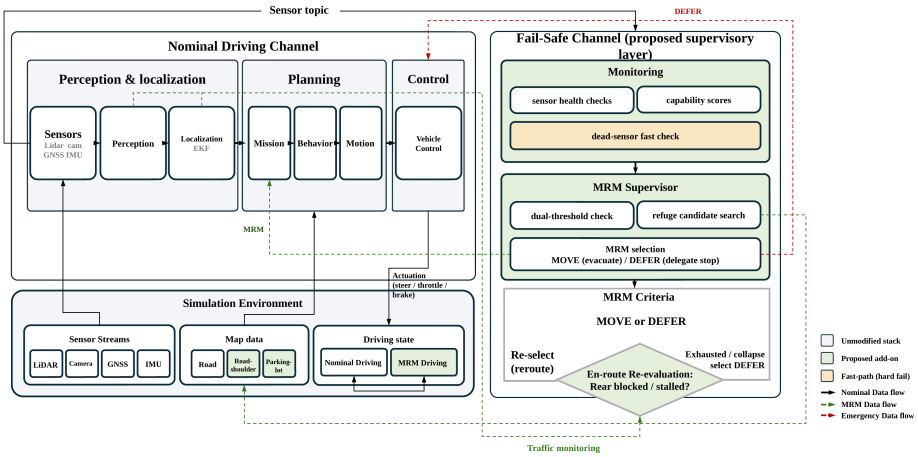


Fig. 1. Dual-channel structure. Left: the nominal driving channel of unmodified Autoware; right: the proposed supervisory layer—monitoring subscribes read-only to sensor topics and produces health/capability scores; the decision stage selects MOVE (evacuate: reroute with a speed cap) or DEFER (inject a diagnostic ERROR to trigger the stock stop), re-evaluating rear occupancy and stalls en route. Bottom: the CARLA closed loop; refuge search draws on annotated map data.

precise threshold placement. Timing splits by evidence type: degradation faults are statistical evidence requiring sustained confirmation (1.0 s band entry; 2.0 s at the floor since deferral is terminal; a 5.0 s recovery grace, unchanged under a 0.5–5.0 s sensitivity sweep), whereas hard fails (silence past the 850 ms freshness deadline) already carry accumulated evidence: confirmations are skipped and unrecoverable conditions defer immediately. Selection is not frozen while moving: sustained rear occupancy demotes reversing parking candidates to a shoulder stop, stalls get bounded retries, and exhaustion or health collapse falls back to DEFER.

Refuge selection minimizes

$$J(c) = w_t\,\eta(c) + w_{tl}\,n_{tl}(c)\,g(tl) + w_{sec}\sum(1 - s_{\text{req}}) + w_{loc}\,\eta(c)(1 - \text{loc}), \quad (1)$$

where $\eta(c)$ is degraded-state exposure time (route time + expected signal waits)—risk accumulates per unit time, not distance. Route times and traffic-light counts are computed on an in-house road-network graph parsed directly from the lanelet2 map, keeping the decision layer outside the influence of main-stack faults. The key safeguard is the traffic-light observability gate: a candidate whose route passes signals no healthy camera can read is excluded with $J = \infty$. Shoulder/parking candidates are manually annotated map attributes (16 shoulders, 4 lots on the test map).

F. Conservative Low-Speed MRM Movement (Speed Cap)

Evacuation is movement under degraded perception, so stopping capability against a dart-out dominates. With reaction time τ and deceleration a, stopping from speed v requires

$$d_{\text{safe}}(v) = v\tau + \frac{v^2}{2a} + d_{\text{margin}} \le D_{\text{avail}}. \quad (2)$$

Here $\tau \approx 1.5$ s, $a \approx 3.5$ m/s^2 are nominal lumped values calibrated to the measured ladder boundary—a design approximation, not a predictive safety model. On entering MRM_MOVE the supervisor imposes a fixed 3.0 m/s cap, chosen inside the band between the model-inverted cap that still stops within an 8 m dart-out (≈3.9 m/s) and the requirement that evacuation-completion time stay bounded; no in-band optimization is claimed. The cap feeds Autoware's velocity-limit arbiter, and delivery is verified by measured vehicle speed, not publish records—an interface mismatch that silently dropped the cap in earlier runs was caught exactly this way and corrected (Sec. V-D).

IV. EXPERIMENTAL SETUP

A. Environment and Campaign Matrix

The closed loop couples CARLA 0.9.16 with Autoware 0.39.0 over the AWCL bridge [17] on Town07 (annotated custom vector map; speed limit 15 km/h); the simulator is fully restarted for every case. A case is {method, fault, severity, traffic, repetition, injection time}; methods are ours, baseline (identical detection, deferral to the stock stop—isolating response strategy from detection), and mrm-off (passive scoring only). The main campaign is fault × severity × traffic density (none/defensive/aggressive) × method with 1,131 valid cases (39 short of plan, all in high-severity LiDAR clutter); a navigation-extension matrix adds four navigation faults (611 valid), and encounter experiments complete the 1,862 total. The fault model injects inside the agent's sensor pipeline: camera death/noise (cf/cn), LiDAR death/sparsification/clutter (lf/lso/lcl, physical models [8], [9]), GNSS noise/correlated bias/outage (gn/gb/gf [10]), and IMU noise/bias/drift (imn/imb/imd), with severities low/mid/high where applicable; injection times and fault waveforms are seed-fixed for pairing. Fig. 2 illustrates the sensor-level effect of each injection.

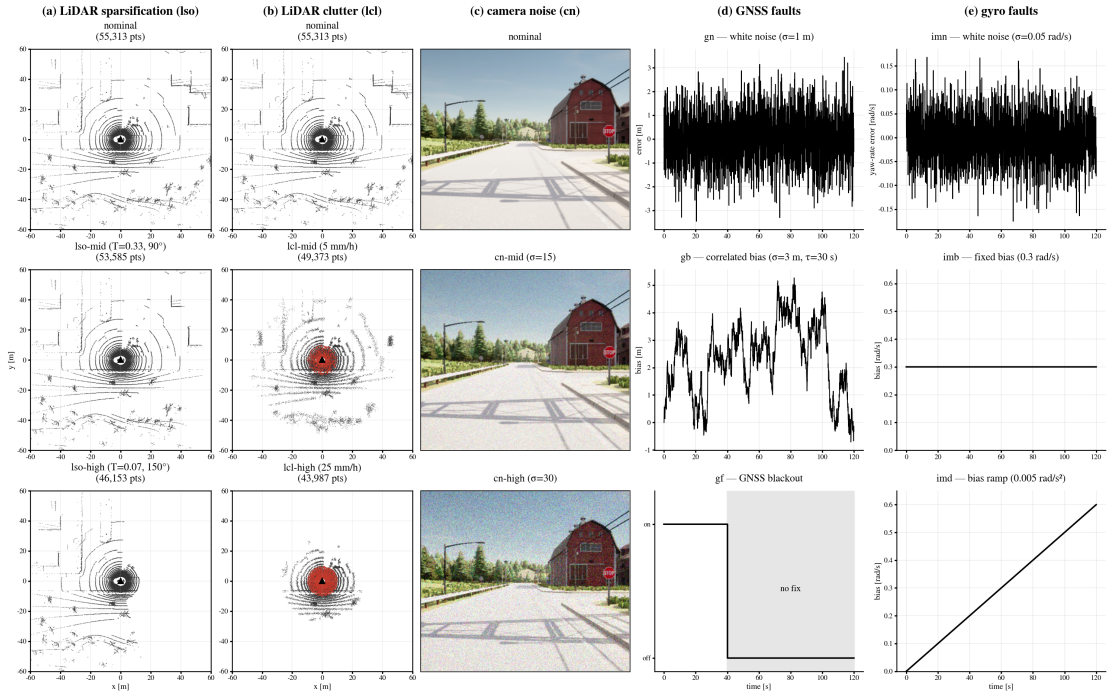


Fig. 2. Sensor-level effects of fault injection—the injector's own physics applied to one measured Town07 LiDAR scan and camera frame. (a) LiDAR sparsification (lso); (b) LiDAR clutter (lcl, spurious returns in red); (c) camera pixel noise (cn); (d) GNSS fault waveforms (gn/gb/gf); (e) gyro fault waveforms (imn/imb/imd). Panel parameters are the injector's raw values for the severity definitions.

B. Collision Adjudication

The initial implementation adjudicated collisions with a collision sensor attached to the pedestrian; measurement showed this sensor never fired in CARLA, and a listener death lost separation records, so non-intervention-arm cases were being misjudged as avoided (affected data were discarded and re-run). The reconstructed system has two layers: (i) primary: a real-time ledger (RAW-HIT) written by the patched callback of the ego's sensor.other.collision the moment a collision occurs, with counterpart actor identifiers—unlike official infraction tallies it survives early/forced termination; (ii) audit: a pedestrian physics oracle (flung >4.5 m/s or a >5 m jump within 1 s) and the per-tick separation stream. Runner exit labels are refused as evidence (endpoint contacts, post-collision stops, and infractions surviving only natural termination). All encounter-level figures here are ledger-based; latencies use simulation time.

C. Encounter Experiments

A decoupled injector process spawns pedestrian dart-outs at the right lane edge with per-seed fixed walking speed, jitter, and fault waveform. Designs: (i) a fault-free distance ladder at 4.2 m/s, spawn distances $\{14, 11, 8, 6, 5\}$ m, n=4 each; (ii) the same ladder blinded (lso-high); (iii) an 8 m encounter comparison under gn-mid (EKF input contaminated, perception intact—so all arms share detection capability); (iv) a low-speed ladder at 3.0 m/s cruise; and (v) a dose-response experiment: caps $\{4.2, 3.6, 3.0\}$ m/s × distances $\{6, 7, 8, 9\}$ m × 10 seeds = 120 runs, launched by a settlement trigger—the pedestrian spawns D m ahead only when the supervisor is in MRM_MOVE and measured speed has stayed in the cap-specific band for 1.0 s—so an encounter under an actually holding cap is guaranteed at trigger level. The primary endpoint (8 m, paired 4.2 vs 3.0 exact McNemar) and a manipulation-check adoption rule (first-tick speed in band ±0.15 m/s, ahead-of-ego geometry, launch within 420 s) were pre-specified; the sample was fixed at 120 adopted runs.

V. RESULTS

A. Detection Performance

Table I summarizes the campaign: detection 96–100% in 19 of 20 conditions—median latencies ∼1.0 s for hard fails (simulation time), 2.1–8.6 s for degradation faults (wall clock)—with zero false activations in 45 fault-free controls. The sole exception, mid-severity LiDAR sparsification (2%), lies by design in the tolerated band (∼55 m effective range remains; the supervisor deliberately continues driving); since tolerance cannot yet be separated from channel insensitivity, we read the table as intervention eligibility. Notable structure: IMU-drift latency is inversely related to ramp slope (8.6 vs 5.4 s)—small faults are caught later but become dangerous later; GNSS-outage latency is bimodal (intended: a vehicle stopped at a red light is not a risk exposure); and deferral stops split by family (GNSS → comfortable, IMU → hard stops).

Behind the binary rates lies the continuous signal (Fig. 3): the decision variable $\min(\text{perception}, \text{cap}_{\text{loc}})$

TABLE I

Detection rate and median latency per fault condition (n=45 unless noted; †: navigation extension; *: simulation time, others wall clock).

Fault	Sev.	Det. rate	Med. lat. (s)
cf camera death	–	1.00	0.98*
cn camera noise	mid/high	0.98/1.00	3.3/3.2
lf LiDAR death	–	1.00	1.00*
lso LiDAR sparsif.	mid/high	0.02/0.98	–[c]/5.0
lcl LiDAR clutter	mid/high[a]	1.00/0.96	4.8/4.5
gn GNSS noise	mid/high	1.00/1.00	2.1/4.4
imn IMU noise	mid/high	1.00/0.98	4.8/3.7
gb† GNSS corr. bias	mid/high	1.00/1.00	3.0/2.3
gf† GNSS outage	–	1.00	3.2 (driving)[b]
imb† IMU gyro bias	mid/high	1.00/1.00	3.6/3.2
imd† IMU bias ramp	mid/high	1.00/1.00	8.6/5.4
nf fault-free control	–	0 false activations / 45	

[a] lcl-high n=26 (final batch halted; accounted).
[b] bimodal: 21–40 s if the outage coincides with a stop.
[c] tolerated band; latency not reported.

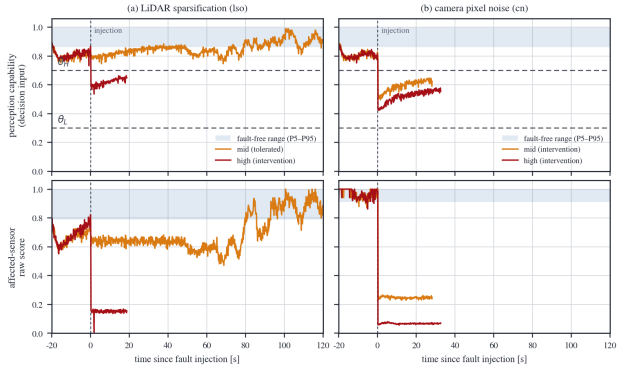


Fig. 3. Injection-aligned health traces (top: the perception capability score feeding the 0.7/0.3 transitions; bottom: raw sensor score). Fault-free stays in the normal band, mid severity in the tolerated band, high severity departs immediately—the band separation underlying the graded response.

separates into distinct bands by fault intensity—fault-free runs never fell below 0.71 after warm-up (consistent with zero false activations), mid severities stay in the tolerated band, and high severities depart immediately on injection—the basis of the graded normal/evacuate/defer structure.

B. Graded Evacuation Behavior and Exposure

For faults preserving the capability to move, the proposal evacuated while the baseline (by definition) stopped in lane: camera death 16/45 shoulder evacuations (27 intended deferrals, 2 maneuver incompletions), camera noise 25/90, GNSS noise 42/90. Severity mattered most in the navigation family: gn-mid produced 40/45 evacuations while gn-high converged to deferral in 42/45—movement is abandoned when localization confidence falls below the movement minimum; conversely lso-mid continued driving deliberately (response proportional to degree). Unmonitored (mrm-off, one run per type) the stack completed 238–275 m faulted; with monitoring, exposure became a policy variable: 32–58 m for detect+evacuate, 9–18 m for detect+stop (11–12 m for LiDAR clutter). The stop minimizes exposure at the cost of lane occupancy; evacuation accepts tens of meters of controlled exposure for lane clearance and an off-lane refuge.

TABLE II

Dose-response: collisions per cell (n=10 seeds, ledger adjudication, settled encounters).

Cap (m/s)	6 m	7 m	8 m	9 m	Boundary
4.2 (control)	10/10	10/10	10/10	10/10	>9 m
3.6	10/10	10/10	10/10	2/10	∼9 m
3.0 (policy)	8/10	5/10	0/10	0/10	∼7 m

C. Avoid/Collide Boundary and the 8 m Encounter

The fault-free ladder at 4.2 m/s produced a clean staircase: 14 and 11 m avoided 4/4 each; 8, 6, 5 m collided 4/4 each—boundary in 8–11 m, matching Eq. (2)'s 8.8 m (the same data calibrated the model—self-consistency). Blinded (lso-high), the stock stack froze mid-road before the encounter at 14/11 m and collided at ≤8 m: an unmonitored degraded vehicle either stands still or strikes.

Under gn-mid (perception intact, so all arms share detection capability), an 8 m dart-out gave: stock at 4.2 m/s collided 10/10; the proposed low-speed settled cruise avoided 8/8 and completed evacuation (auxiliary Fisher $p = 2.3 \times 10^{-5}$); and—critically—the same policy collided 2/2 in the deceleration transient (3.5–3.8 m/s): protection begins only after deceleration completes. The low-speed ladder placed the 3.0 m/s boundary at 7–8 m (8 m: 8/8 avoided; 7/6/5 m: 2/2 collisions each).

D. Dose-Response Verification of the Speed Cap

Because the two ladders used different launch protocols, the cap effect was verified in a controlled preregistered experiment on settled encounters only, after fixing a cap-delivery interface defect (the earlier "capped" runs were receiving no cap—publish records are not delivery; measured speed is). All 120 adopted runs passed the manipulation check (per-arm first-tick speeds 4.05–4.13, 3.37–3.58, 2.60–3.07 m/s; five failing runs rerun blind to outcome). Results (Table II) are monotone along both axes. At the pre-specified primary endpoint (8 m), cap 4.2 collided in all ten seeds and cap 3.0 in none—discordant pairs 10:0, exact McNemar two-sided $p = 0.002$: once the cap actually holds, its collision-reduction effect is detected with unanimous seed-level directionality. The boundary moves inward monotonically (Table II), corroborating the low-speed ladder's 7–8 m interval. Interpretation limits: the causal conclusion is restricted to the settled regime (the cap creates the post-settling margin, not transient protection); no optimality is claimed for 3.0 m/s; cells are n=10, single fault, single corridor.

E. Hard-Fail Decision Latency

Hard fails are decided by an evidence-aware fast path (skipping noise-oriented debounce and futile recovery holds) against an internal 1.5 s design deadline (not an FTTI-compliance claim). Over the full camera/LiDAR-death matrices, ledger-based simulation-time latencies gave medians ≈1.0 s, maximum 1.20 s, zero violations, with MRM attainment maintained in full.

VI. DISCUSSION AND LIMITATIONS

Stop vs. evacuation is a temporal trade-off. The in-lane stop showed the lowest collision rate even under transient-window launches, at the price of sustained lane occupancy; evacuation accepts controlled moving exposure and residual transient risk but on success clears the lane to an off-lane refuge. Neither is unconditionally superior; the choice should follow the fault's temporal structure—concrete selection variables for ISO 23793-1.

Residual risk is temporally localized. All evacuation-arm collisions occurred before deceleration completed (transient 2/2), none after settling (0/8). Distinguishing the post-detection transient from the overall fault-response risk window ($T_{\text{detect}} + T_{\text{decel}}$), the results support—observationally, not causally—that earlier detection and faster deceleration shrink this window, widening the FTTI margin.

Adjudication lessons generalize. "Zero collisions" is meaningful only after the verdict channel is proven alive: positive-control checks of collision sensors, physics-oracle cross-checks, and refusal to read absence of a record as absence of an event. The silently undelivered cap carries the same lesson—actuation must be verified by downstream effect; the real-time ledger was the remedy.

Limitations. Single map, single 15 km/h limit, two launch corridors: the numeric boundaries and dose-response values are low-speed-ODD numbers; generalizable claims are structural (boundary's stopping-distance dependence, transient risk concentration, the stop/evacuation trade-off). Autoware observations are version-specific (0.39.0). Dose-response cells are n=10 with wide CIs; testing was restricted to the pre-specified endpoint. Campaign-level collision-free status predates the ledger (no pedestrian injections there). The dynamic detection budget was not ablated against a fixed deadline; hazard-time-referenced compliance needs an instrumentation extension; the health-proportional cap is implemented but unevaluated; no real-vehicle validation.

VII. CONCLUSION

We proposed and validated, in an Autoware–CARLA closed loop of 1,862 valid cases, a supervisory add-on layer estimating sensor faults as continuous health scores and selecting MRMs in a graded manner, with a conservative cap on evacuation movement. Three conclusions carry beyond the numbers. First, shoulder evacuation is no panacea: stopping and evacuation form a temporal trade-off between immediate collision risk and lane clearance, and the choice should follow the fault's temporal structure. Second, evacuation's residual risk concentrates in the post-detection transient, while settled low speed verifiably moves the boundary inward (transient 2/2 vs. settled 0/8; dose-response 10:0, $p = 0.002$)—supporting earlier detection and faster deceleration within the FTTI. Third, simulation safety claims hold only on a validated adjudication framework; the ledger/physics-oracle scheme and catalogued pitfalls are reusable for any simulated MRM comparison. Future work: high-speed ODDs and maps, formalizing the detection budget, the health-proportional cap, and real-vehicle validation.

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